INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, KURNOOL

Department of Electronics and Communication Engineering

Electronic Prototyping and Manufacturing (EPM) Laboratory

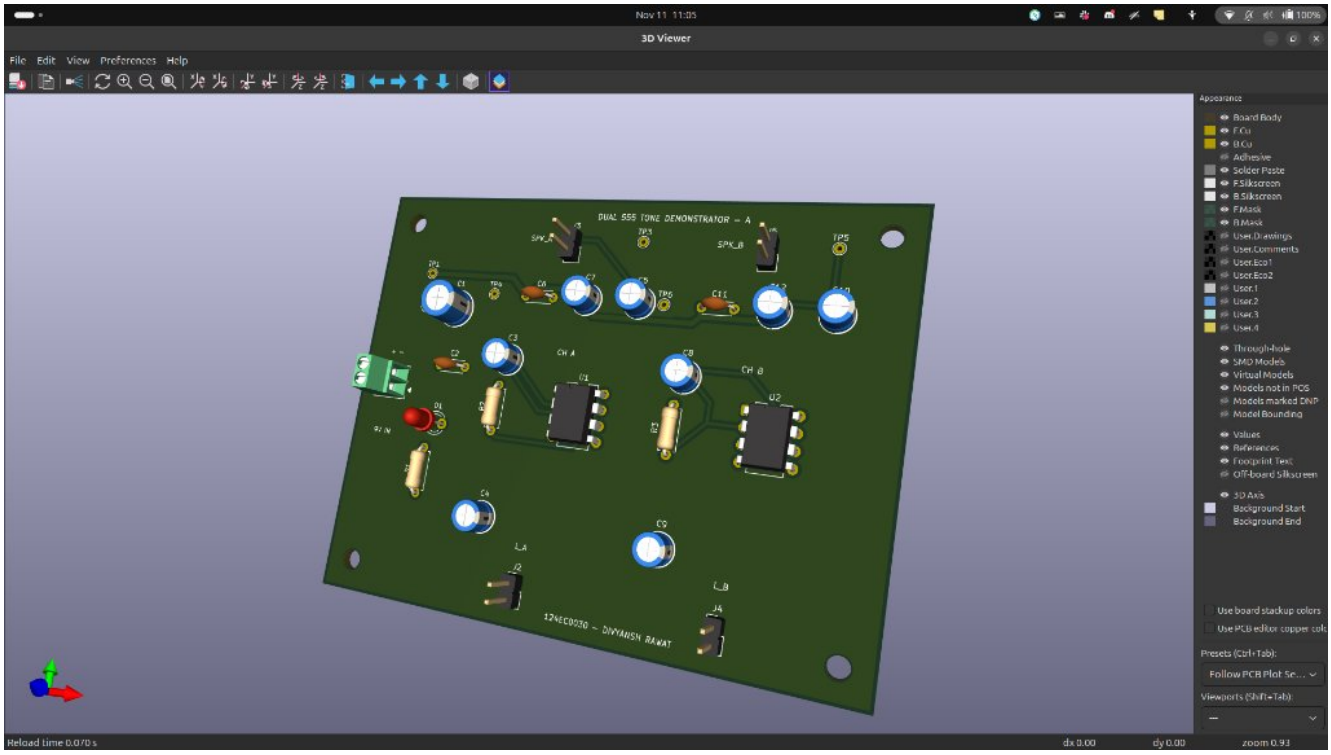
End Semester Practical Examination — Laboratory Report

Student Name:	Divyansh Rawat	**Roll Number:**	124EC0030
Course Faculty:	Dr. Eswaramoorthy K V	**Date of Exam:**	10-11-2026
Project Title:	Dual Inductance-Sensitive Tone Demonstrator - A	**Exam Pattern:**	Layout Variant A (90 x 70 mm)
Design Environment:	KiCad 9.0.8 (x86_64 Linux)	**DRC / ERC Status:**	**0 Errors / 0 Violations (100% Pass)**

EXAM QUESTION SPECIFICATIONS (VARIANT A):

Implement two copies of the source 555 inductance-sensitive oscillator. Use an external inductor connector and speaker connector for each channel. Supply: 9 V DC. Use a 90 x 70 mm rectangular board. Place the supply connector on the left edge, input/user controls along the lower edge and outputs along the upper edge. Keep inductive/timing loops short; label L A/B and SPK A/B connector polarity where applicable. Mandatory circuit clarifications: No search-coil winding, detection-distance target or field calibration is required. This is a layout-only oscillator demonstrator, not a validated general metal detector. Use exact capacitor connections from the schematic. Assign verified through-hole footprints and complete a two-copper-layer PCB using both F.Cu and B.Cu.

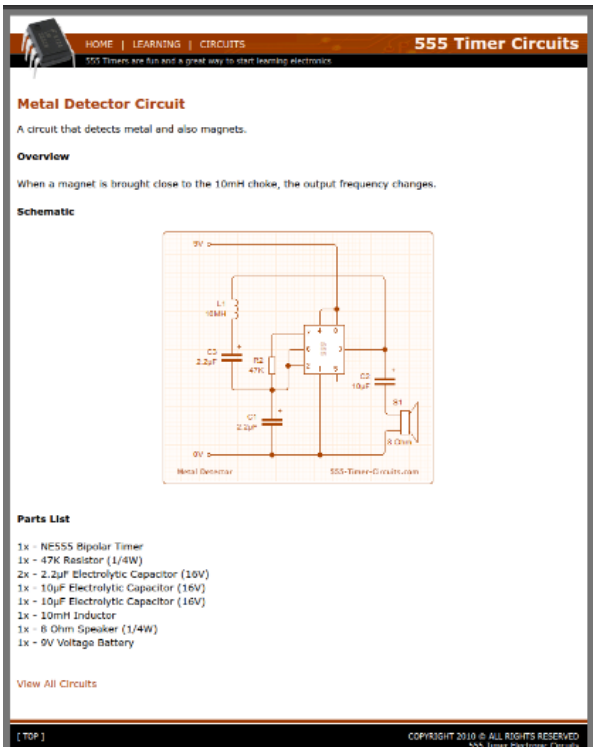
3D BOARD PREVIEW (REALISTIC GREEN SOLDERMASK):



1. CIRCUIT DIAGRAM (REFERENCE CIRCUIT SPECIFICATION)

Source Reference: **555-Timer-Circuits.com Metal Detector / Dual Inductance-Sensitive Tone Generator**. The circuit operates as an astable multivibrator where oscillation frequency is modulated by the inductive reactance of choke L1 in conjunction with timing capacitors and feedback resistor R2. When a metallic object is brought into proximity with the external choke, the inductance shifts, producing an audible frequency change.

Source Website & Parts List:



HOME | LEARNING | CIRCUITS

555 Timer Circuits

555 Timers are fun and a great way to start learning electronics

Metal Detector Circuit

A circuit that detects metal and also magnets.

Overview

When a magnet is brought close to the 10mH choke, the output frequency changes.

Schematic

Parts List

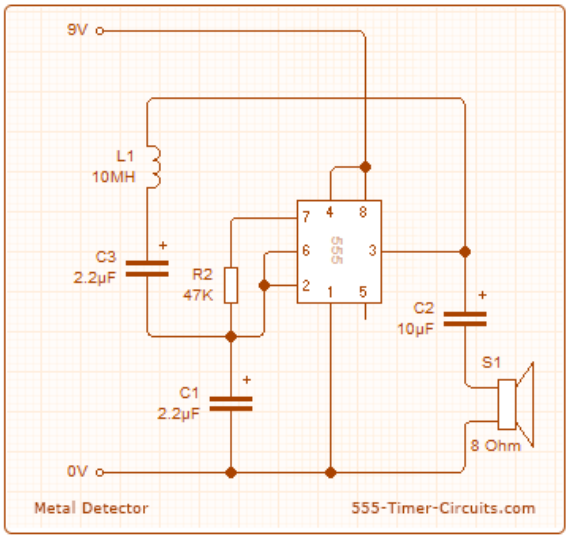
1x - NE555 Bipolar Timer
1x - 47K Resistor (1/4W)
2x - 2.2uF Electrolytic Capacitor (16V)
1x - 10µF Electrolytic Capacitor (16V)
1x - 10µF Electrolytic Capacitor (16V)
1x - 10mH Inductor
1x - 8 Ohm Speaker (1/4W)
1x - 9V Voltage Battery

View All Circuits

[TOP]

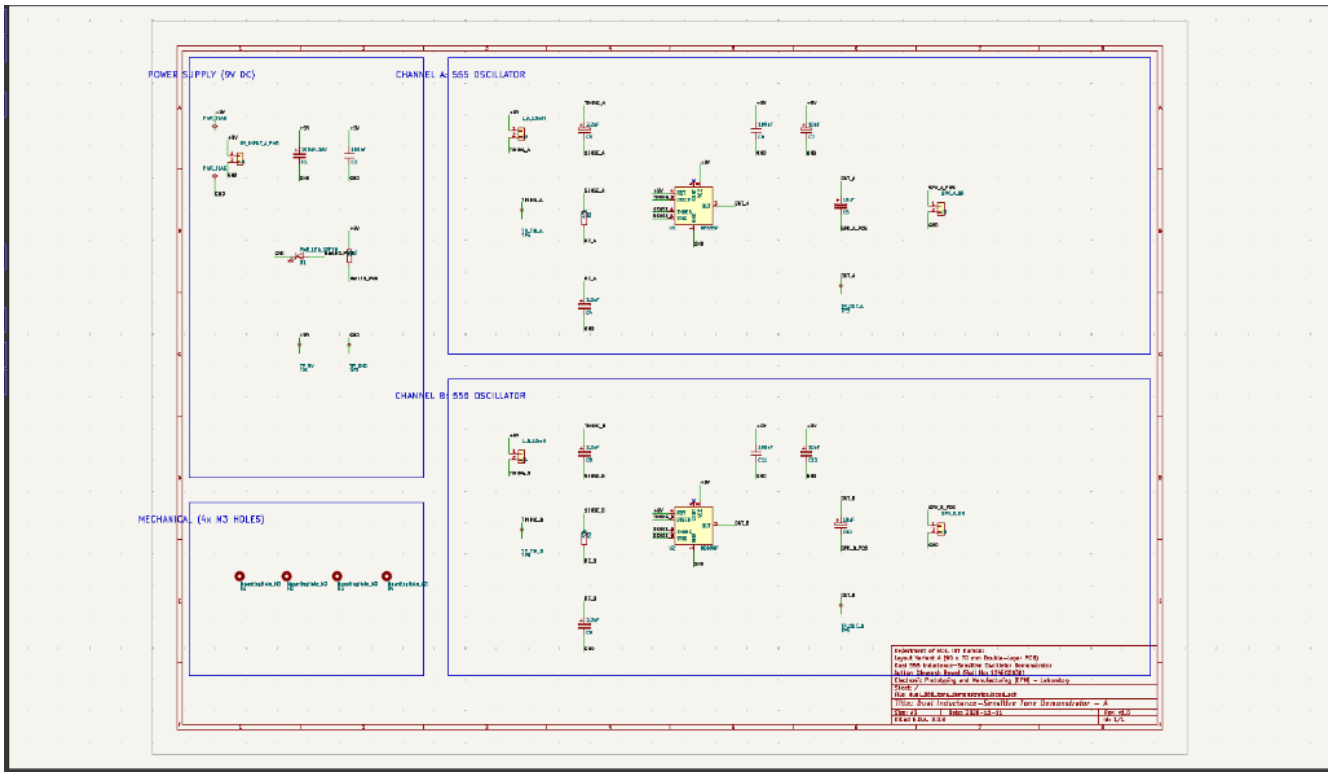
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Enlarged Circuit Schematic:



2. SCHEMATIC DIAGRAM (KiCad Schematic Editor)

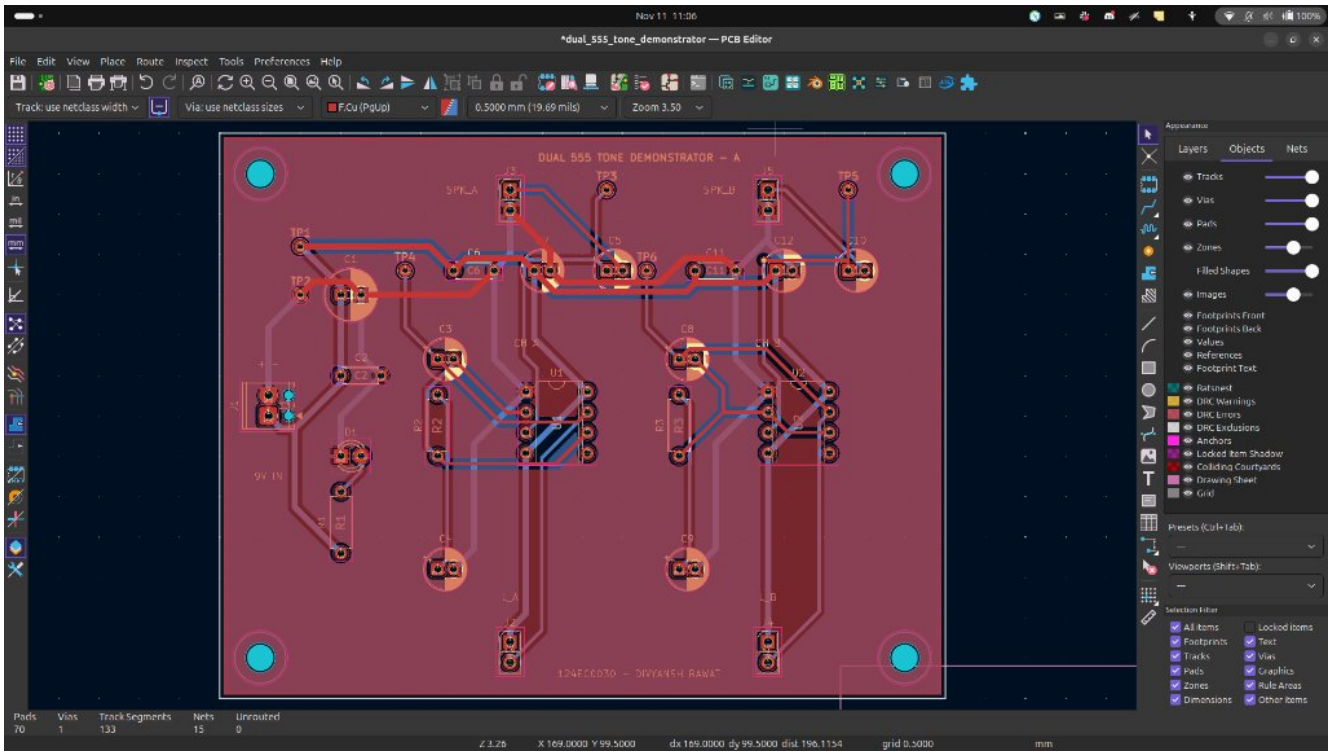
Live interactive KiCad Schematic Editor capture showing the complete dual 555 oscillator design. The schematic is organized into four distinct functional modules: (1) 9V DC Power Supply Conditioning with reverse polarity protection and bulk/high-frequency filtering; (2) Channel A 555 astable oscillator with external inductor connector J2 and speaker connector J3; (3) Channel B 555 astable oscillator with external inductor connector J4 and speaker connector J5; and (4) Mechanical M3 chassis mounting holes (H1-H4).



Sheet Size / Frame:	A3 (420 x 297 mm, Standard Drawing Frame)	**Hierarchy:**	Flat Single Sheet (Root /)
Oscillator Channels:	2 Identical 555 Astable Circuits (Ch A & Ch B)	**Decoupling Capacitors:**	100 nF Ceramic per IC (Pin 8 to GND)
ERC Status:	**0 Errors, 0 Warnings (Passed)**	**Schematic File:**	dual_555_tone_demonstrator.kicad_sch

3. COMPOSITE PCB LAYOUT (KiCad PCB Editor)

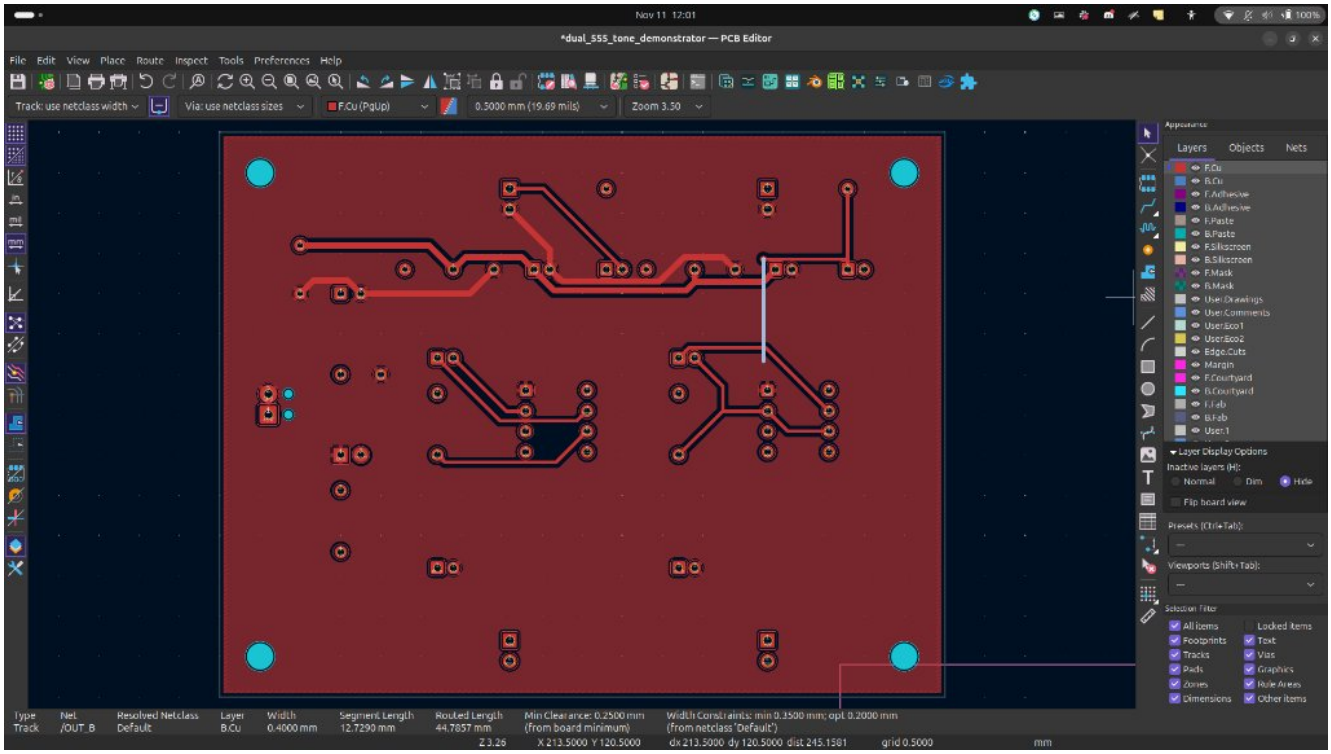
Live interactive PCB Editor screenshot showing double-sided routing (F.Cu red, B.Cu blue), continuous solid ground planes with thermal relief spokes, component placements, and Edge.Cuts outline. Supply connector J1 is placed on the left edge; external inductor headers J2 & J4 along the lower edge; external speaker headers J3 & J5 along the upper edge; timing loops kept minimal; 4x M3 mounting holes at corners.



Board Dimensions:	90.00 mm x 70.00 mm (Rectangular)	**Layer Stackup:**	2 Copper Layers (F.Cu & B.Cu, 1.6 mm FR4)
Component Count:	23 Through-Hole Components	**Routing Completion:**	**49 / 49 Nets (100% Routed, 0 Unrouted)**
Mounting Holes:	4x M3 (3.2 mm NPTH, 6.0 mm Keepout)	**Diagnostic Test Points:**	6 THT Pads (TP1 - TP6)

4. FRONT COPPER LAYER (F.Cu)

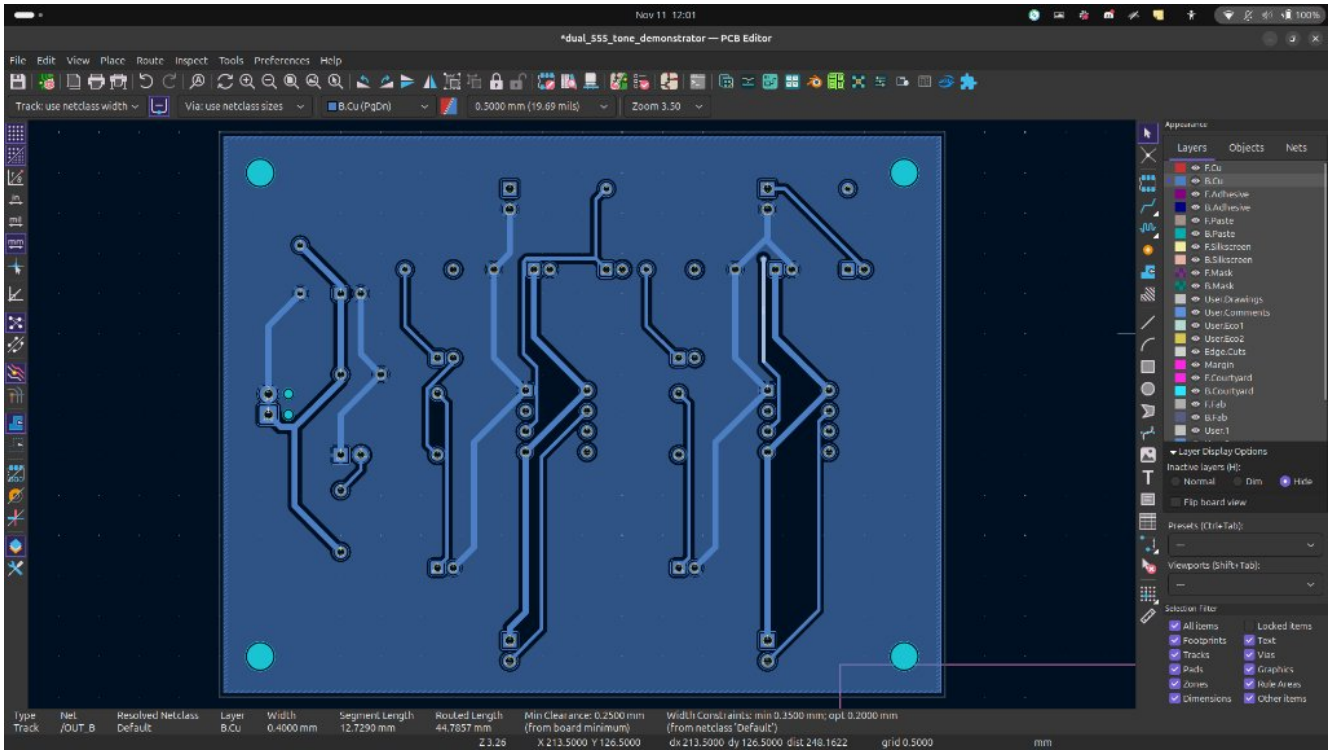
Live KiCad PCB Editor layer inspection view showing the top copper routing (F.Cu) with all inactive layers hidden. Features 0.50 mm wide power distribution tracks (+9V DC rail) and 0.30 mm signal tracks. All trace corners utilize 45-degree mitered chamfers (0.60 mm setback) to prevent 90-degree acid traps during chemical etching.



Layer Name:	F.Cu (Top Copper, Red)	**Copper Thickness:**	35 µm (1.0 oz/ft² finished)
Power Track Width:	0.50 mm (19.69 mils, Net /+9V)	**Signal Track Width:**	0.30 mm (11.81 mils)
Minimum Clearance:	0.25 mm (Board Minimum)	**Corner Miter Geometry:**	45° Diagonal Transitions

5. BACK COPPER LAYER (B.Cu)

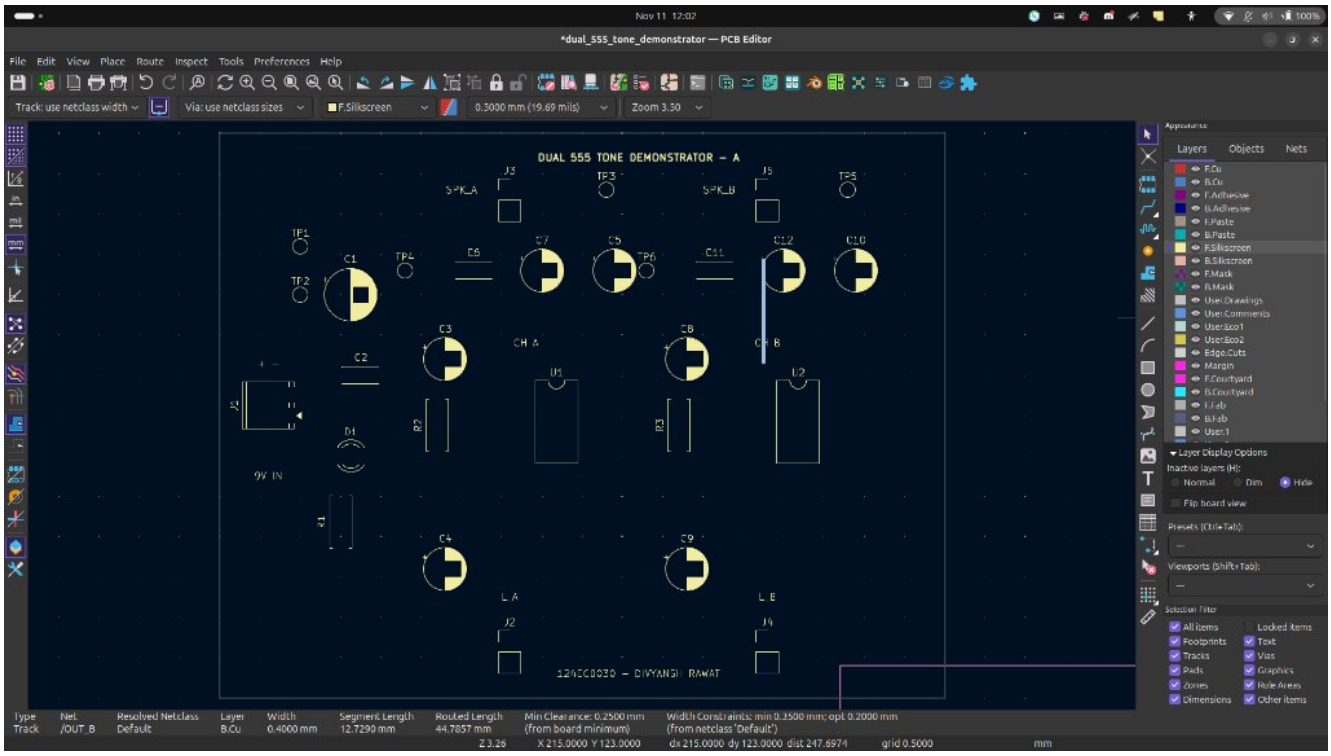
Live KiCad PCB Editor layer inspection view showing the bottom copper routing (B.Cu) with all inactive layers hidden. Provides an expansive, low-impedance ground return plane with thermal relief spoke connections to through-hole GND pads. Interconnects Channel A and Channel B oscillator timing sub-circuits while preserving return current continuity.



Layer Name:	B.Cu (Bottom Copper, Blue)	**Copper Thickness:**	35 µm (1.0 oz/ft² finished)
Ground Plane Pour:	Solid Continuous Zone (Net /GND)	**Thermal Relief Spokes:**	4 Spokes (0.50 mm width, 0.35 mm gap)
Unconnected Copper:	0 Isolated Copper Islands	**Corner Miter Geometry:**	45° Diagonal Transitions

6. SILKSCREEN LAYER (F.SilkS)

Live KiCad PCB Editor layer inspection view showing the front silkscreen legend (F.SilkS) with all inactive layers hidden. Contains unambiguous component outlines, reference designators (U1-U2, R1-R3, C1-C12, D1, J1-J5, TP1-TP6), polarity indicators (+/- for 9V IN, pin 1 notches for DIP-8 ICs, and electrolytic capacitor positive markings), connector labels (9V IN, L_A, L_B, SPK_A, SPK_B), and student roll number identification.

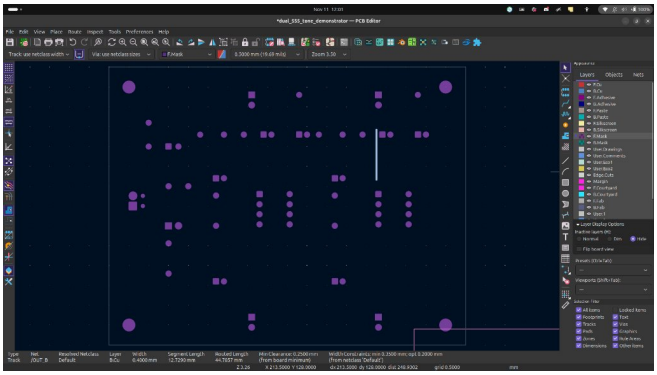


Layer Name:	F.SilkS (Front Silkscreen, Yellow/White)	**Silkscreen Color:**	White on Green Solder Mask
Text Height / Width:	1.0 mm / 1.5 mm (0.15 mm line thickness)	**Polarity Labels:**	9V IN (+/-), L A/B, SPK A/B, C1-C12 (+)
Student Attribution:	124EC0030 — DIVYANSH RAWAT	**Title Legend:**	DUAL 555 TONE DEMONSTRATOR - A

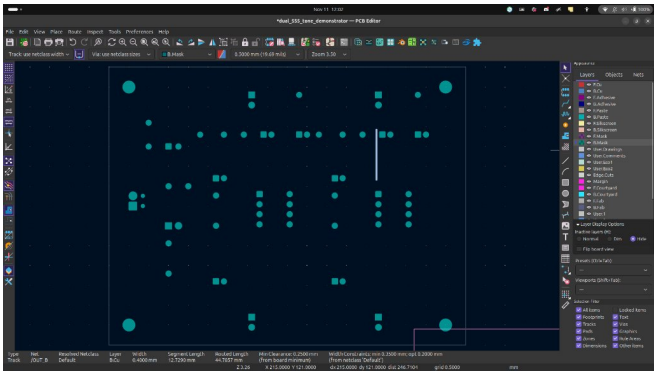
7. SOLDER MASK LAYER (F.Mask & B.Mask)

Live KiCad PCB Editor layer inspection views showing solder mask apertures with inactive layers hidden. Top apertures (F.Mask, purple) and bottom apertures (B.Mask, teal) provide 0.10 mm radial clearance around all through-hole component pads, test points, terminal blocks, and mounting holes to prevent solder bridging during hand soldering.

Front Solder Mask Apertures (F.Mask - Top Side):



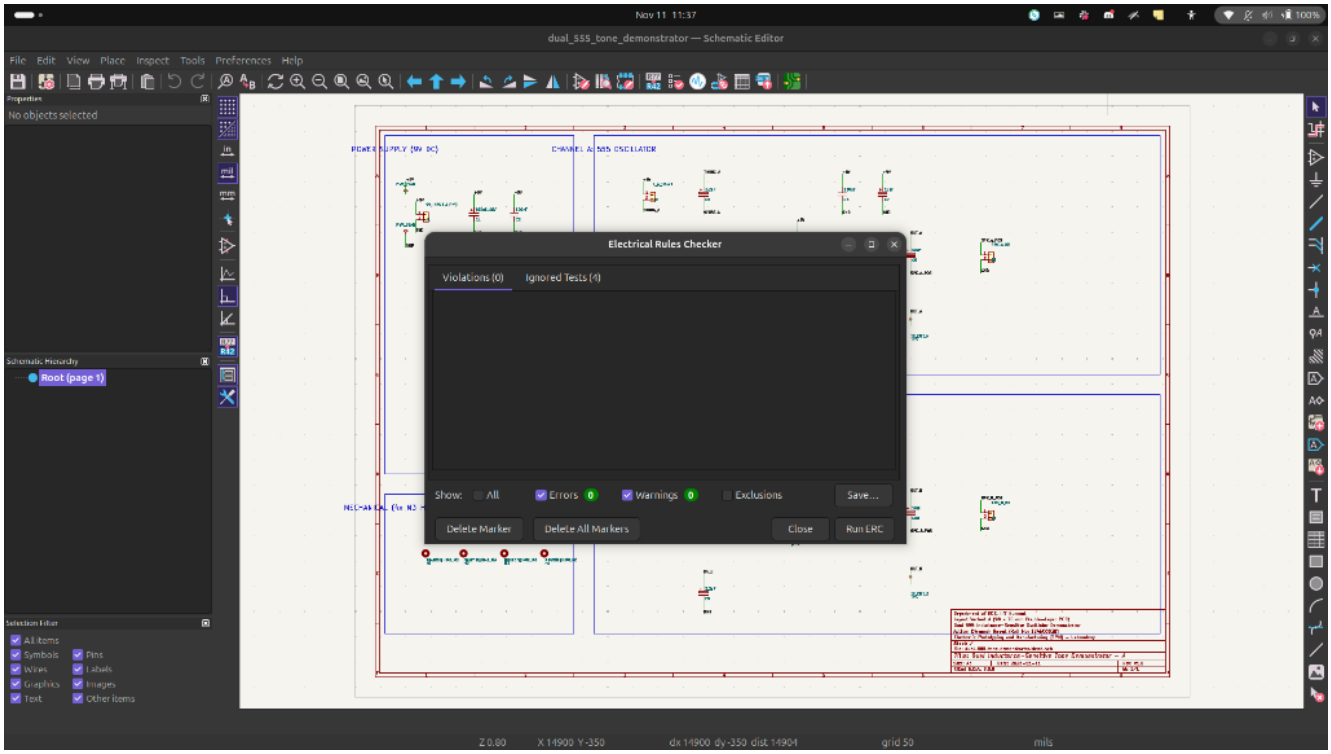
Back Solder Mask Apertures (B.Mask - Solder Side):



Mask Clearances:	0.10 mm (Board Design Rules)	**Minimum Solder Dam:**	0.10 mm Between Adjacent Pads
Pad Aperture Coverage:	All 70 THT Component & Connector Pads	**Mask Finish / Color:**	Liquid Photoimageable (LPI) Green

8. SCREENSHOT OF ELECTRICAL RULES CHECK (ERC)

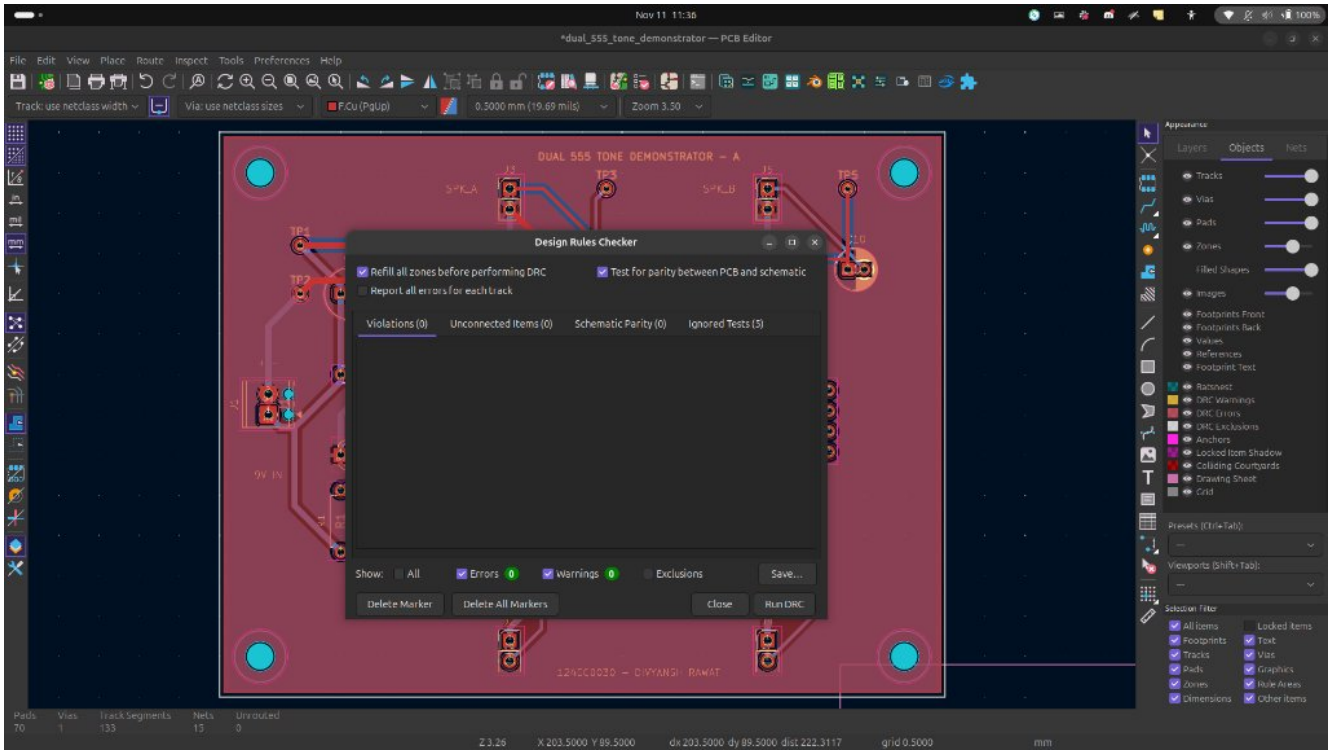
Live KiCad Schematic Editor Electrical Rules Checker dialog showing **0 Errors, 0 Warnings, and 0 Violations**. Confirms that all pin electrical types, power flag drivers, net connectivity, and label references match design rules.



ERC Status:	**PASSED (0 Errors, 0 Warnings)**	**KiCad Version:**	9.0.8+dfsg-1 (Linux x86_64)
Schematic File:	dual_555_tone_demonstrator.kicad_sch	**Sheet Name:**	Root Sheet (/)
Power Flags:	PWR_FLAG configured on +9V & GND	**Unconnected Pins:**	0 Unintended Unconnected Pins

9. SCREENSHOT OF DESIGN RULES CHECK (DRC)

Live KiCad PCB Editor Design Rules Checker dialog showing **0 Violations, 0 Unconnected Items, and 0 Schematic Parity Warnings**. Confirms 100% routing completion, strict clearance compliance, zero courtyard overlaps, and full board-to-schematic netlist synchronization.

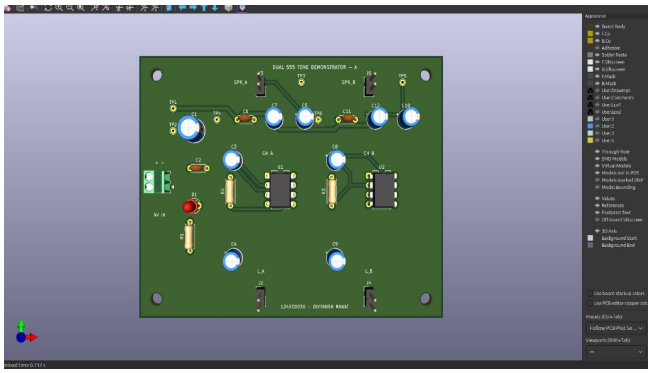


DRC Violations:	**0 Violations**	**Unconnected Items:**	**0 Unconnected Items**
Schematic Parity:	**0 Issues (100% Match)**	**Overall DRC:**	**0 Errors, 0 Warnings**
Zone Refill Status:	All Zones Filled Prior to Check	**Board File:**	dual_555_tone_demonstrator.kicad_pcb

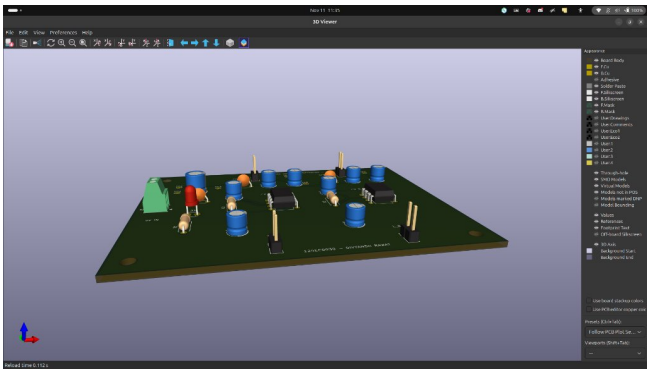
APPENDIX A: 3D VISUALIZATION VIEWS (REALISTIC GREEN SOLDERMASK)

Comprehensive 3D renders from four vantage points demonstrating component package fit, through-hole solder joints, connector orientations, and ground planes.

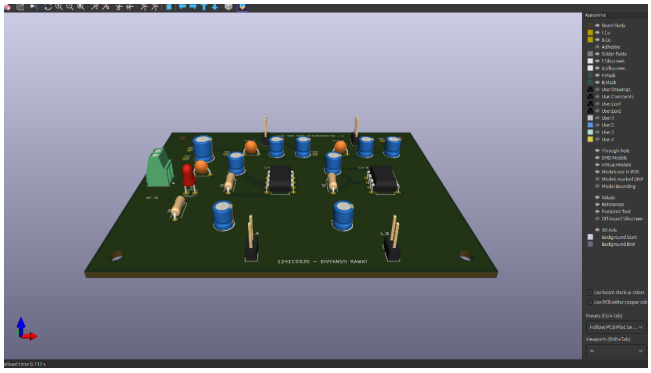
Top Orthogonal View:



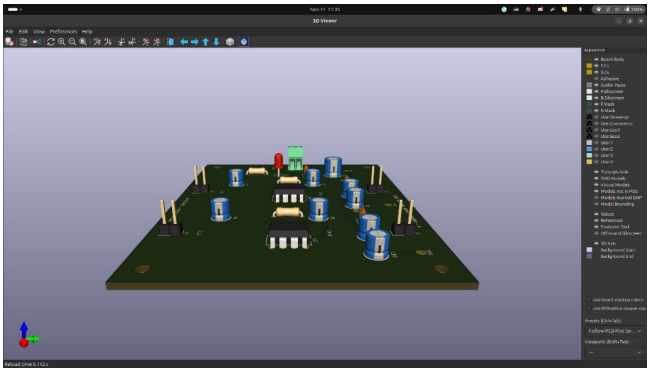
Isometric Perspective View:



Front Elevation View:



Side Profile View:



APPENDIX B: BILL OF MATERIALS (BOM)

Item	Reference	Value	Footprint	Qty	Description / Function
1	U1, U2	NE555P	DIP-8_W7.62mm	2	Precision Bipolar Timer IC
2	J1	9V_INPUT	TerminalBlock_MPT-0.5-2_P2.54mm	1	DC Power Input Terminal Block
3	J2, J4	L_A, L_B	PinHeader_1x02_P2.54mm_Vertical	2	External Inductor (10 mH) Connectors
4	J3, J5	SPK_A, SPK_B	PinHeader_1x02_P2.54mm_Vertical	2	External Speaker (8 Ohm) Connectors
5	C1	100uF_16V	CP_Radial_D6.3mm_P2.50mm	1	Power Supply Bulk Reservoir Capacitor
6	C2, C6, C11	100nF	C_Disc_D4.3mm_W1.9mm_P5.00mm	3	High-Frequency Ceramic Bypass Capacitors
7	C3, C4, C8, C9	2.2uF_16V	CP_Radial_D5.0mm_P2.00mm	4	Astable Timing Electrolytic Capacitors
8	C5, C7, C10, C12	10uF_16V	CP_Radial_D5.0mm_P2.00mm	4	AC Audio Output Coupling Capacitors
9	R1	1k	R_Axial_DIN0207_P7.62mm	1	Power LED Current-Limiting Resistor
10	R2, R3	47k	R_Axial_DIN0207_P7.62mm	2	Astable Timing Feedback Resistors
11	D1	PWR_LED	LED_D3.0mm	1	Green Power Status Indicator
12	TP1-TP6	TestPoint	TestPoint_THTPad_D1.5mm	6	Diagnostic Test Points (+9V, GND, Outs, Timings)
13	H1-H4	M3 Hole	MountingHole_3.2mm_M3	4	M3 Mechanical Chassis Mounting Holes